

A
Major Project
On
**PRIVACY ENHANCED ANONYMOUS CLOUD
AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR
PROTECTION**

(Submitted in partial fulfilment of the requirements of the award of Degree)

BACHELOR OF TECHNOLOGY

**In
INFORMATION TECHNOLOGY**

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CERTIFICATE

This is to certify that the project entitled “**PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION**” being submitted by **P Pravallika (227R1A12B8), R Narendar (227R1A12C2) & Ch Sai Kiran (227R1A1275)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

Cloud computing offers cost-effective, scalable storage and computation, making it the preferred choice for social networking platforms, enterprises, and individual users. However, despite the convenience, the centralized nature of cloud services introduces significant risks to user data privacy and security. While existing solutions incorporate encryption to safeguard stored data, they often fail to provide user anonymity or protection against impersonation attacks, two critical vulnerabilities that can enable insider threats or unauthorized access. This work proposes a privacy-preserving cloud storage framework that addresses these gaps by integrating identity anonymization, session-based authentication, and hybrid lightweight encryption. To preserve user anonymity, the framework hashes usernames and biometric images during registration, ensuring that even the cloud server cannot identify or track user activities. To prevent cloud or user impersonation attacks, the system establishes secure sessions at login, which are immediately terminated at logout, preventing post-session misuse of access credentials. The framework's workflow includes user registration, where usernames and biometric data are hashed before storage; user login/verification, which matches entered credentials and biometric data with hashed records; secure file upload, where files are encrypted using Elliptic Curve Cryptography (ECC) for robust security, supplemented by CHACHA, a lightweight encryption algorithm offering over 50% faster computation; and file download, enabling only verified users to access and decrypt data. Additionally, a computation graph module compares the performance of ECC and CHACHA, illustrating the efficiency gains from using the hybrid encryption approach. This system ensures end-to-end security through anonymization, strong encryption, and controlled access. Testing confirms that the framework effectively conceals user identities from cloud administrators, mitigates impersonation risks, and improves encryption performance without compromising data confidentiality. This approach advances cloud security by combining cryptographic robustness with privacy-focused identity management, making it suitable for environments where data sensitivity and resistance to insider threats are paramount. The solution is adaptable for integration into multi-tenant cloud architectures, e-governance services, and secure enterprise collaboration platforms.

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1. INTRODUCTION

1. INTRODUCTION

The rapid growth of cloud computing has enabled organizations to store and access data efficiently through scalable and remote digital platforms, but it has also introduced significant security and privacy challenges in user authentication. Traditional cloud systems mainly rely on single-factor login methods such as usernames and passwords, which are vulnerable to cyber threats like credential theft, phishing, brute-force attacks, and identity impersonation. Additionally, centralized authentication processes often expose real user identities during login, increasing the risk of privacy leakage and unauthorized tracking. To address these issues, this project proposes a *Privacy-Enhanced Anonymous Cloud Authentication Framework with Multi-Factor Protection* that integrates strong cryptographic techniques including SHA-256 hashing, ChaCha20 encryption, and Elliptic Curve Cryptography. The system ensures secure credential storage, encrypted communication, and reliable identity verification within cloud environments. It also introduces anonymous identity tokens that preserve user privacy while maintaining authentication accountability. Furthermore, multi-factor authentication mechanisms such as password validation, token verification, and session monitoring strengthen access control. By combining privacy preservation, layered security verification, and automated authentication processes, the framework enhances trust, scalability, and protection against modern cloud authentication threats.

1.1 PROJECT PURPOSE

Several research works have focused on improving cloud authentication security using cryptographic techniques and multi-factor verification mechanisms. Researchers have explored various methods such as public-key encryption, hashing algorithms, and secure token-based authentication to protect user identities and prevent unauthorized access in distributed cloud environments. One study presented an authentication scheme based on Elliptic Curve Cryptography that provides strong security with reduced computational overhead, making it suitable for scalable cloud systems.

Modern cloud security solutions also emphasize privacy preservation during authentication. Some researchers have proposed anonymous credential systems that replace real user identities with temporary identifiers to reduce the risk of identity tracking and data leakage. Additionally, multi-factor authentication models combining password verification with token

validation or device confirmation have shown significant improvement in preventing credential theft and brute-force attacks.

With the increasing adoption of cloud computing across industries, there is a growing need for secure authentication frameworks that ensure confidentiality, integrity, and availability of user data. This project aims to develop a privacy-enhanced anonymous authentication system that integrates cryptographic security mechanisms and layered verification strategies to provide reliable access control and strengthen trust in cloud service platforms.

1.2 PROJECT FEATURES

The proposed system includes several advanced features that enable secure and privacy-preserving cloud authentication:

1. Secure Credential Processing Using Cryptographic Techniques

The system protects sensitive authentication data by applying SHA-256 hashing and ChaCha20 encryption before storing credentials in the cloud database. This ensures strong data integrity and prevents attackers from accessing raw passwords or authentication information even if the database is compromised.

2. Elliptic Curve Cryptography Based Key Generation

The framework uses Elliptic Curve Cryptography to generate public-private key pairs for secure communication and identity verification. This cryptographic mechanism provides high security with lower computational cost, making it suitable for real-time cloud authentication processes.

3. Anonymous Identity Authentication

A key feature of the system is the generation of anonymous identity tokens that replace real user identifiers during authentication sessions. This privacy-preserving approach prevents identity exposure while still allowing the cloud server to validate legitimate users.

4. Multi-Factor Authentication Mechanism

PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION

One of the main features of the project is its ability to classify cyber threats into different categories. For example, it can identify whether a threat is related to malware, phishing, ransomware, or other types of attacks. This classification helps users quickly understand the nature of the threat.

5. Secure Session Management

The framework monitors user sessions after successful authentication to prevent session hijacking or misuse. Automatic timeout and logout mechanisms help maintain continuous protection of cloud resources.

6. Role-Based Access Control

The system assigns access privileges based on user roles such as administrator or normal user. This ensures that authenticated users can only access authorized cloud resources, improving overall system security.

7. Real-Time Security Monitoring

The authentication framework tracks login attempts, failed authentications, and suspicious activities to generate security alerts when abnormal patterns are detected. This proactive monitoring helps in early threat detection and mitigation.

8. Web-Based User Interface

The project is implemented as a Django-based web application that provides a simple and user-friendly interface for registration, login, and authentication management. This makes the system accessible even to users with limited technical expertise.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

Mostafa et al. developed an adaptive multi-factor and multi-layer authentication framework for cloud platforms, aiming to significantly enhance security and reduce false positive alarms for unauthorized access. They implemented a dynamic authentication approach that utilized several factors for identity verification, which included examining the length and validity of the user's credential, along with checks based on the user's geolocation and browser confirmation method. This framework achieved a more robust and adaptive access control system and employed intrusion detection mechanisms; however, the proposed work did not explicitly address the critical requirement of user anonymity or privacy-preserving techniques during the authentication process.

Alatawi et al. They researched critical challenges in cloud security by proposing a novel framework that integrated blockchain-based smart contracts to enhance authorization and authentication processes. The authors leveraged smart contracts to enable decentralized, transparent, and tamper-proof mechanisms for managing access control in cloud environments. The proposed system mitigated prevalent threats like unauthorized access and identity theft by providing an immutable and auditable security framework. A prototype system, developed using the Ethereum blockchain and Solidity programming, demonstrated the feasibility of this decentralized security approach, although the work did not specify the use of multi-factor protection or anonymous credentialing.

Tanveer et al. presented in CMAF-IoT is built on the ASCON authenticated encryption (AE) system, which combines encryption and decryption with authentication to provide secrecy, integrity, and authenticity. As a result, designing an authentication framework requires fewer cryptographic procedures.

Alasmary presented in ESCI-AKA was applied using a Scyther tool with the use of a random oracle model and informal security analysis. Furthermore, the analysis of ESCI-AKA and other renowned security systems demonstrates that it has minimal computational and communication overhead, while offering strong security issues.

Qui et al. presented a 3FA protocol is applied to provide secure, efficient, and practical for mobile lightweight devices. The extended chaotic maps component of the protocol is used to generate random numbers. The fuzzy verifier's component of the protocol is used to verify the users' identity.

Alsirhani et al. The main methodology is based on applying different layers of authentications to verify cloud users and reduce false alarms. Furthermore, different methods must be applied to check cloud user identity and maintain the secrecy of data. The key threats in different cloud computing applications and environments include data loss, hijacking of accounts, malicious users, and leakage of data.

Wang et al. They defined for explaining failures in multi-factor authentication (MFA). One of these factors is the incomplete definition of an adversary, in which the capabilities and goals of an attacker and difficulties in defining cryptographic primitives must be defined. In addition, the provided MFA frameworks may be complex or unable to identify vulnerabilities. These factors are checked with eight proof failures to examine vulnerabilities.

Wu, L. et al. They researched on securing the encryption key in remote data backup by proposing a User-Centric Design (UCD) scheme based on multi-factor authentication. Recognizing that remote channels and backup servers are untrustworthy, the scheme employed a secret sharing mechanism to divide the encryption key into three parts, which were securely stored across the user's laptop, smart card, and the server. This key could then be reconstructed using any two of these shares, combined with the user's private information, including their password, identity, and biometrics. The design successfully resisted various threats, such as replay, impersonation, and offline password guessing attacks, provided the biometrics possessed sufficient entropy.

Hu et al. found that the design of Liu et al.'s scheme in the authentication phase is unreasonable, as their scheme cannot resist offline password-guessing attacks, server/user camouflage attacks, and so on. Then, Hu et al. presented an enhanced secure data backup scheme to overcome all above-mentioned security threats.

2.1 REVIEW OF RELATED WORK

Several researchers have contributed to improving cloud authentication security using cryptographic techniques and multi-factor verification mechanisms. Li et al. proposed a privacy-preserving cloud authentication scheme based on elliptic curve cryptography that enables secure identity verification while reducing computational overhead. Their work highlighted the importance of lightweight cryptographic methods in distributed cloud environments. Similarly, Wang et al. introduced a secure multi-factor authentication model

that integrates password-based login with token verification to prevent unauthorized access and credential theft in cloud platforms.

Chen et al. developed an anonymous authentication framework that uses dynamic identity generation techniques to protect user privacy during remote login sessions. Their approach demonstrated that anonymous credentials can significantly reduce the risk of identity tracking and profiling. In another study, Zhang et al. presented a blockchain-supported authentication mechanism for decentralized cloud systems, ensuring transparency and tamper-resistant access control. These studies collectively emphasize the need for scalable authentication models that combine encryption, anonymity, and layered verification strategies to address evolving cloud security challenges.

2.2 DEFINITION OF PROBLEM STATEMENT

Cloud computing environments handle massive volumes of sensitive user data and authentication requests, but traditional authentication mechanisms rely mainly on single-factor login methods such as username and password verification. These approaches are highly vulnerable to attacks including credential theft, phishing, brute-force attempts, replay attacks, and identity impersonation. Additionally, centralized authentication systems often expose real user identities during login processes, leading to privacy concerns and increased risk of user tracking or data leakage. Existing authentication frameworks also lack dynamic risk assessment capabilities and fail to provide secure encrypted communication during credential transmission. As cloud services expand across multiple platforms and devices, the need for scalable and privacy-preserving authentication becomes more critical. To address these challenges, the proposed system introduces a privacy-enhanced anonymous cloud authentication framework that integrates SHA-256 hashing, ChaCha20 encryption, Elliptic Curve Cryptography, and multi-factor authentication mechanisms.

2.3 EXISTING SYSTEM

The existing cloud authentication systems are primarily based on traditional login mechanisms that depend on static credentials and centralized identity verification processes. In such systems, users authenticate themselves using a username and password combination,

which is verified by a central server before granting access to cloud resources. Although this approach is simple and widely used, it introduces significant security vulnerabilities.

Many traditional systems store user credentials in databases with minimal encryption or rely on outdated hashing mechanisms, making them susceptible to data breaches. Moreover, authentication communication channels are not always fully protected, allowing attackers to intercept login requests and perform replay or man-in-the-middle attacks. Another major limitation is the lack of privacy protection, as real user identities are transmitted during authentication, increasing the risk of identity exposure.

Disadvantages

1. Traditional authentication systems rely mainly on single-factor verification such as passwords, which are easy to guess, steal, or reuse across platforms, leading to increased security risks.
2. Centralized identity storage creates a single point of failure, making systems vulnerable to large-scale data breaches and unauthorized credential access.
3. Static authentication mechanisms cannot adapt to evolving cyber-attack strategies such as session hijacking, credential stuffing, or phishing-based intrusions.
4. Lack of anonymous identity protection exposes user information during login sessions, increasing privacy concerns and potential misuse of personal data.
5. Limited monitoring of authentication behavior results in delayed detection of suspicious login attempts or insider threats.

2.4 PROPOSED SYSTEM

To overcome the limitations of traditional authentication approaches, this project proposes a privacy-enhanced anonymous cloud authentication framework with multi-factor protection. The system is designed to provide secure identity verification while preserving user privacy and ensuring encrypted communication during authentication processes.

In the proposed framework, user credentials are first processed using SHA-256 hashing to ensure secure storage and data integrity. The hashed credentials are further protected using ChaCha20 encryption before being stored in the cloud database. During authentication,

Elliptic Curve Cryptography is used to establish secure communication channels and validate user identity through cryptographic key verification.

The system also introduces anonymous identity tokens that replace real user identifiers during login sessions. This mechanism prevents identity tracking while maintaining accountability within the cloud infrastructure. Additionally, multi-factor authentication is implemented by combining password verification, cryptographic token validation, and secure session monitoring. These layered security measures significantly enhance authentication reliability and protect against unauthorized access attempts.

Advantages of the proposed system

1.Enhanced Security Protection: The system improves cloud security by integrating cryptographic encryption, hashing, and multi-factor authentication mechanisms to prevent unauthorized access.

2.Privacy Preservation: Anonymous identity tokens protect real user identities during authentication, reducing risks of identity tracking and data exposure.

3.Secure Credential Storage: User credentials are protected using SHA-256 hashing and ChaCha20 encryption, ensuring strong data integrity and resistance to database breaches.

4.Multi-Layer Authentication Verification: Sequential validation of password, cryptographic token, and session parameters increases authentication reliability and reduces cyber-attack risks.

2.5 OBJECTIVES

- To design and develop a privacy-preserving authentication framework for secure cloud access control.
- To implement SHA-256 hashing and ChaCha20 encryption techniques for protecting user credentials and authentication data.
- To integrate Elliptic Curve Cryptography for secure key generation and encrypted communication between users and cloud servers.
- To develop an anonymous identity generation mechanism that prevents identity exposure during authentication sessions.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

The system requires appropriate hardware and software resources to ensure efficient execution

2.6.1 HARDWARE REQUIREMENTS

- **Processor:** Intel Core i3/i5 or higher (or equivalent AMD processor)
- **RAM:** Minimum 4 GB (8 GB recommended for better performance)
- **Storage:** At least 100 GB free disk space (SSD preferred for faster access)
- **Network Connectivity:** Stable internet connection for cloud authentication operations
- **Display:** Minimum Screen Resolution 1366×76.

2.6.2 SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10/11 or Linux (Ubuntu recommended)
- **Programming Language:** Python 3.7.x
- **Web Framework:** Django
- **Frontend Tools:** HTML, CSS, JavaScript
- **Database:** MySQL or SQLite
- **Web Browser:** Google Chrome / Microsoft Edge / Firefox
- **Cryptographic Libraries:** hashlib (SHA-256), ecies / cryptography (ECC), ChaCha20 modules

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

3.1 PROJECT ARCHITECTURE

The proposed system is a secure authentication architecture developed for the *Privacy-Enhanced Anonymous Cloud Authentication Framework with Multi-Factor Protection*, focusing on strengthening access control and preserving user privacy in cloud environments. It begins with secure user registration where credentials are protected using SHA-256 hashing for integrity and encrypted with ChaCha20 before being stored in the cloud database, while Elliptic Curve Cryptography (ECC) generates cryptographic key pairs to enable secure identity verification and encrypted communication between users and cloud servers. A key innovation of the framework is the use of anonymous identity tokens that replace real user identities during authentication, ensuring privacy without compromising verification accuracy. During login, the system performs sequential multi-factor validation including password authentication, cryptographic token verification, and session confirmation to ensure only legitimate users gain access.

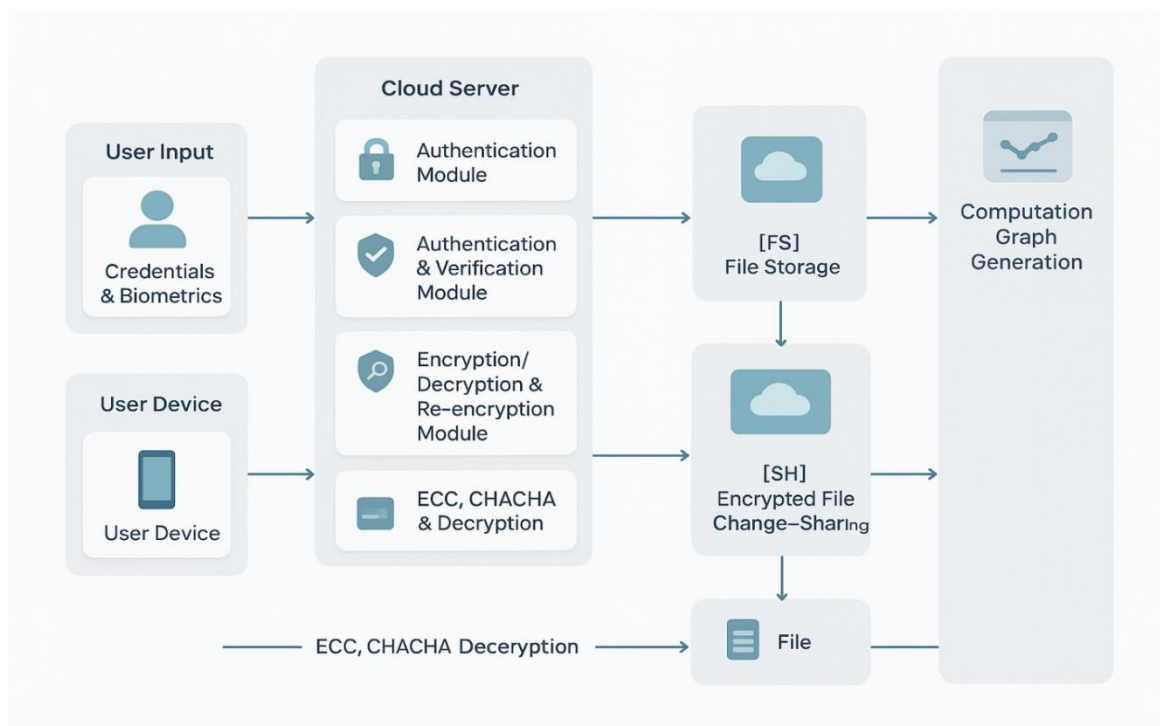


Figure 3.1 Proposed System Architecture of Privacy Enhanced Anonymous Cloud Authentication Framework with Multi Factor.

3.2 DESCRIPTION

1. User Registration and Secure Credential Processing

The system initiates with the user registration module, where authentication details such as username and password are collected through a secure web interface. Before storage, the password is processed using the SHA-256 hashing algorithm to ensure data integrity and prevent exposure of raw credentials. The hashed credentials are then encrypted using ChaCha20 and stored in the cloud database along with system-generated anonymous identifiers.

2. Cryptographic Key Generation and Secure Communication Setup

In this stage, Elliptic Curve Cryptography (ECC) is used to generate public-private key pairs for each registered user. The public key is used for secure communication and authentication validation, while the private key is securely maintained by the system. This cryptographic infrastructure enables encrypted data exchange between users and cloud services, ensuring confidentiality and protection against network-based attacks.

3. Anonymous Identity Generation and Token Management

To enhance privacy protection, the system generates temporary anonymous identity tokens that replace real user identities during authentication sessions. These tokens are dynamically mapped to actual user credentials within the system backend. This approach prevents identity tracking while maintaining accountability and traceability for authorized cloud operations.

4. Multi-Factor Authentication Verification and Session Management

During login, the authentication request passes through multiple verification layers including password validation, token verification, and session confirmation. The system ensures that access is granted only when all authentication factors are successfully validated. Secure session management mechanisms monitor login duration, detect suspicious activities, and automatically terminate inactive sessions to prevent unauthorized access.

5. Access Control Enforcement and Security Monitoring

After successful authentication, role-based access control policies are applied to regulate user permissions within the cloud environment. The system continuously monitors login attempts, failed authentications, and abnormal access patterns to generate security alerts when necessary. This final stage ensures that the framework not only authenticates users securely but also maintains continuous protection against evolving authentication threats.

3.3 UML DIAGRAM

3.3.1 Class Diagram

The class diagram represents the main components of the system and their relationships. It includes User, Admin, File, Authentication, Encryption, Database, and Graph Generation classes. The diagram shows the attributes and methods for each class, highlighting how user registration, login, file upload/download, and encryption operations are structured.

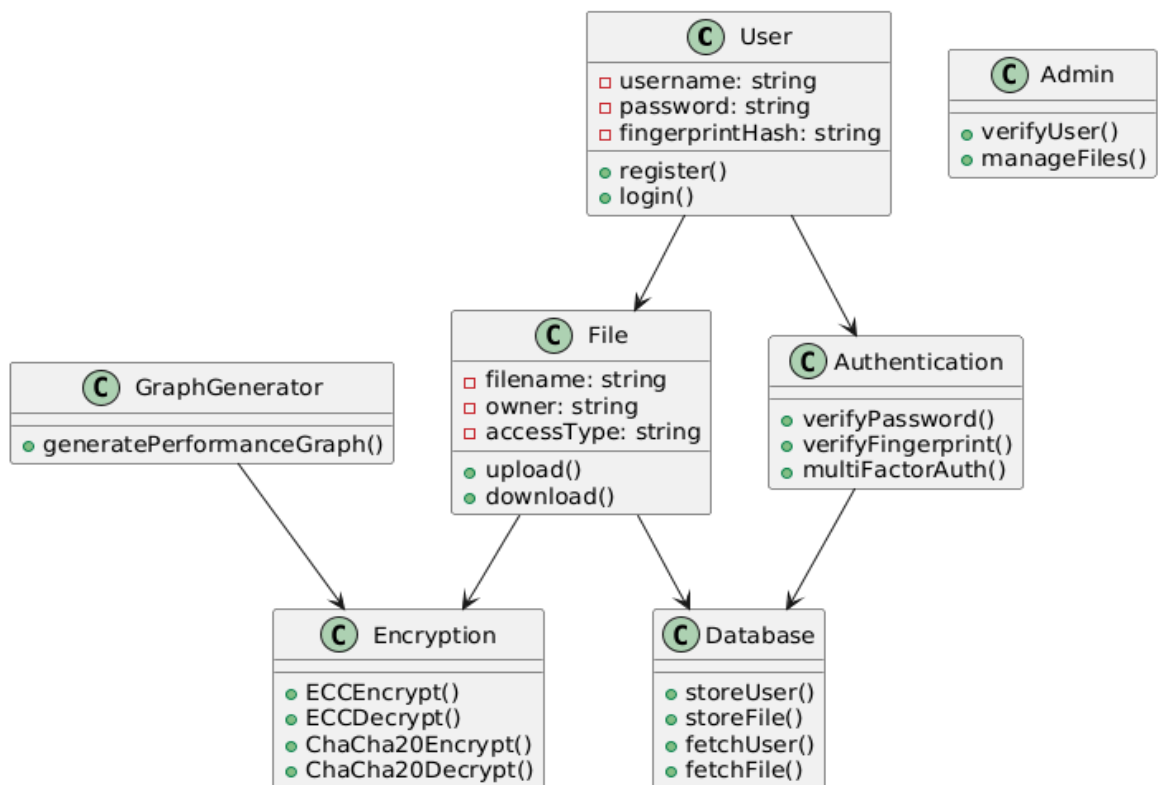


Figure 3.3.1: Class Diagram of Privacy Enhanced Anonymous Cloud Authentication Framework with Multi Factor.

3.3.2 Use Case Diagram

This use case diagram illustrates the interactions between actors (User, Admin) and the system. Key use cases include user registration, login, multi-factor authentication, file upload/download, encryption, and performance visualization. This diagram provides a high-level overview of system functionality.

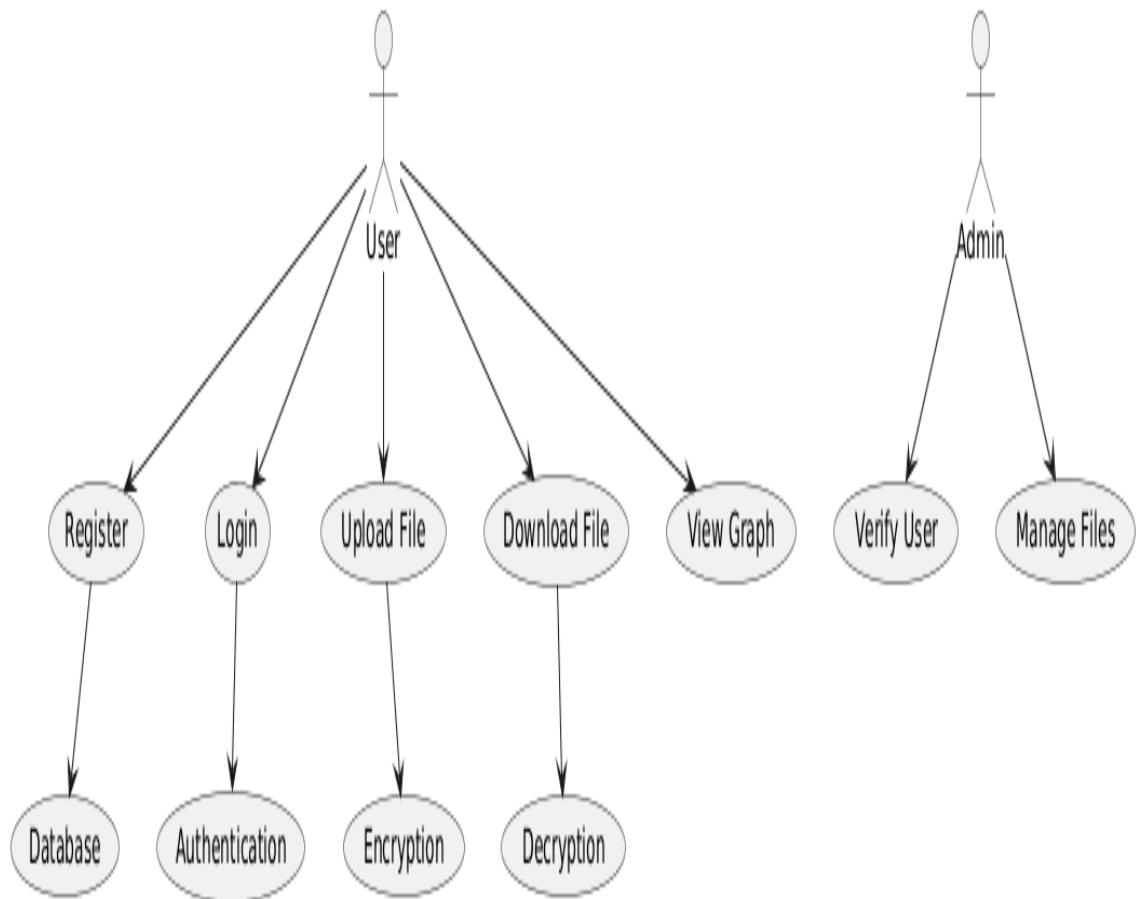


Figure 3.3.2: Use Case Diagram of Privacy Enhanced Anonymous Cloud Authentication Framework with Multi Factor.

3.3.3 Sequence Diagram

The sequence diagram shows the step-by-step flow of operations during file upload and download. It highlights interactions between User, Authentication, Encryption, Database, and File storage, showing the method calls and responses in order.

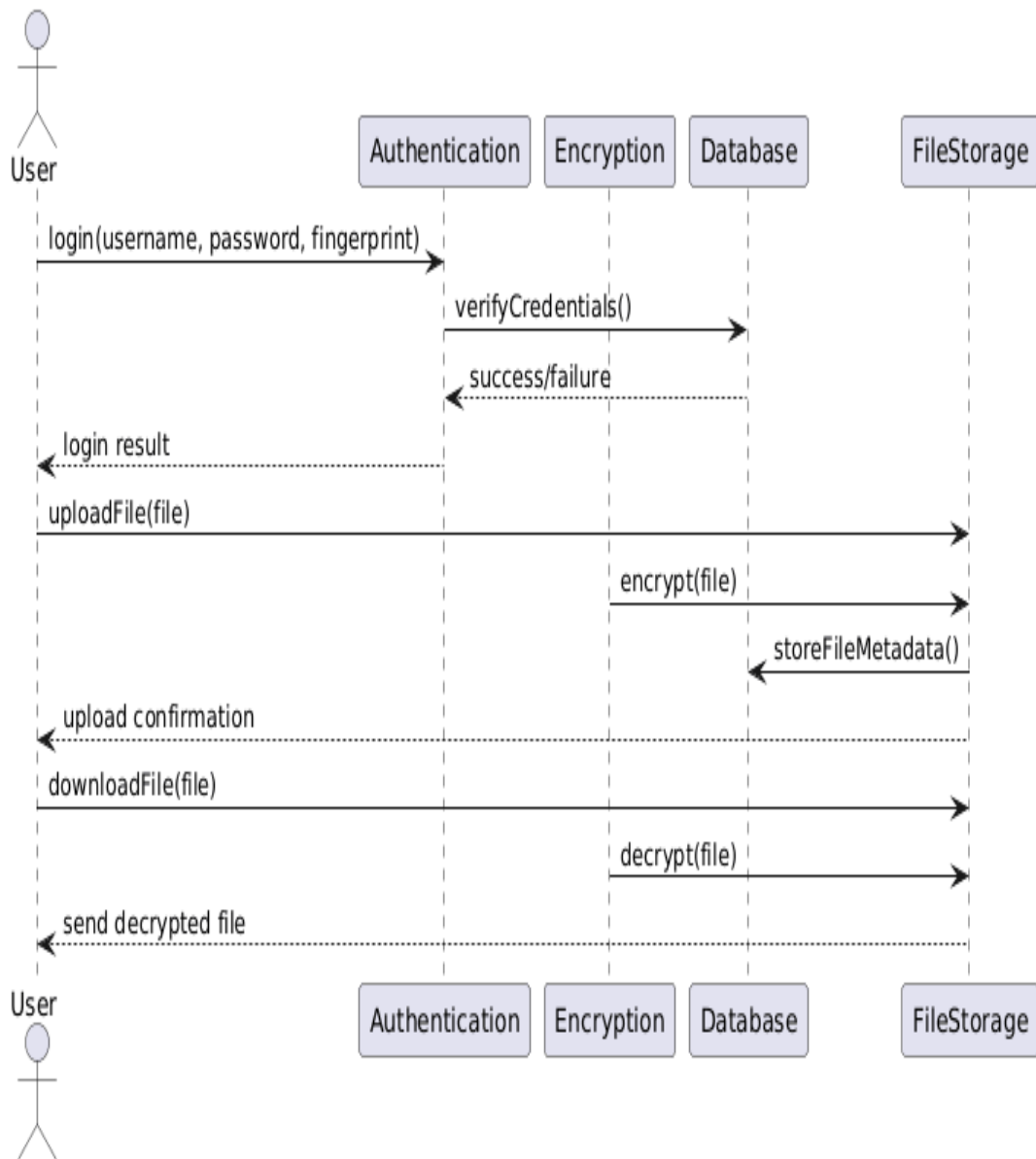


Figure 3.3.3: Sequence Diagram of Privacy Enhanced Anonymous Cloud Authentication Framework with Multi Factor.

3.3.4 Activity Diagram

The activity diagram represents workflow of user file upload, including steps like login, authentication, encryption, storing file, and confirmation. It highlights decision points and sequential operations for clarity.

PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION

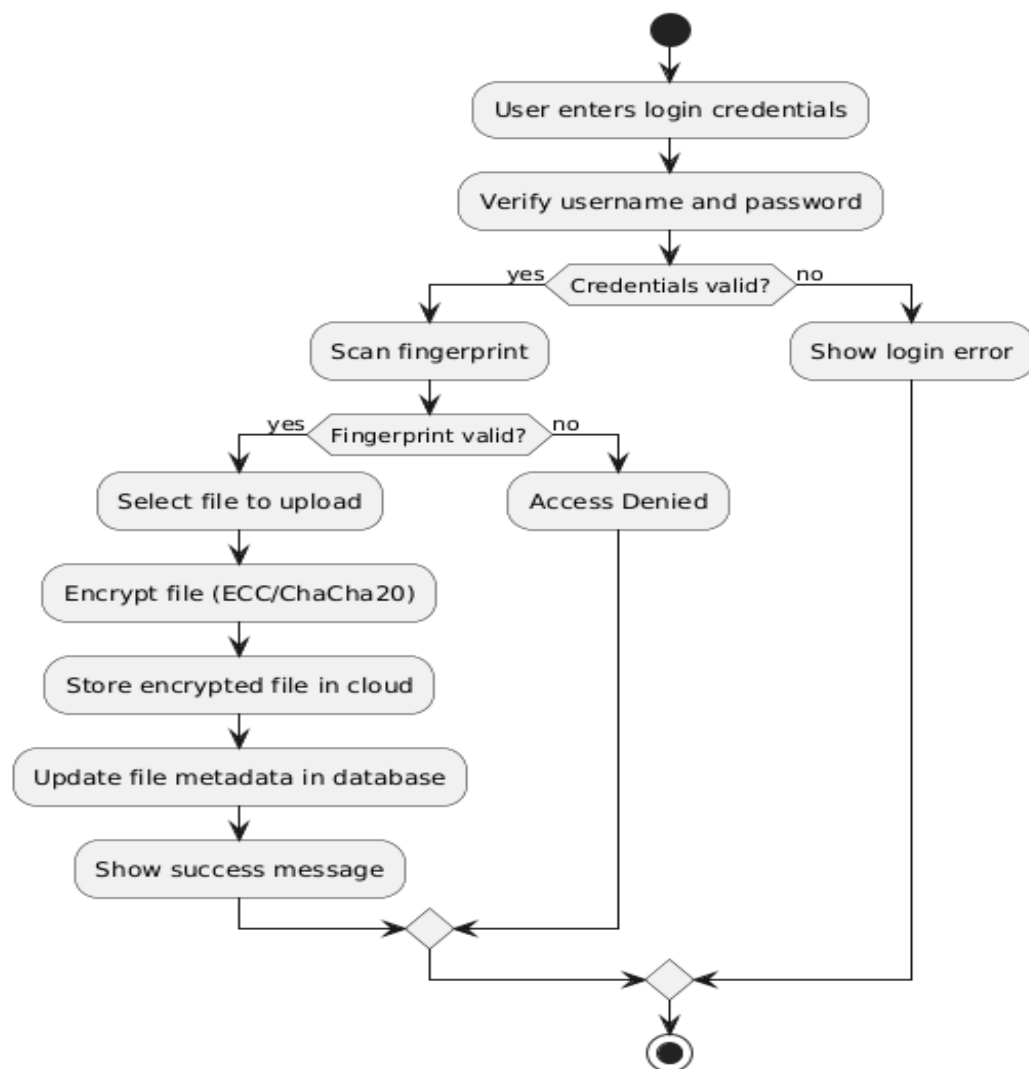


Figure 3.3.4 Activity Diagram of Privacy Enhanced Anonymous Cloud Authentication Framework with Multi Factor.

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation of this project focuses on developing a secure authentication system that ensures privacy-preserving access control in cloud computing environments using cryptographic techniques and multi-factor verification mechanisms. The system is designed as a web-based application so that users can conveniently access cloud services while maintaining strong security and anonymity.

The implementation process begins with user registration and secure credential management. During registration, users provide authentication details which are processed by the system to generate anonymous identity tokens and cryptographic keys. Since authentication data is highly sensitive, it cannot be stored in plain text. Therefore, encryption techniques are applied to protect user credentials before storing them in the cloud database.

After secure data storage, the next stage involves anonymous authentication and encrypted communication setup. In this project, cryptographic methods are used to establish secure channels between users and cloud servers. These mechanisms ensure that user identity remains hidden during authentication while still allowing the system to verify legitimacy. The encrypted communication process helps prevent threats such as eavesdropping, replay attacks, and credential interception.

Once secure communication is established, the multi-factor authentication process is executed. Different authentication factors such as password verification, cryptographic token validation, and session confirmation are applied sequentially. These steps collectively enhance system security by ensuring that access is granted only after successful verification of multiple independent authentication parameters.

The authentication framework is implemented using a Django-based web application that integrates user interfaces, cryptographic modules, and access control logic. When a user attempts to log in, the system processes the request through all authentication stages and provides outputs such as access approval, authentication risk status, and session initiation. This structured implementation enables secure cloud access while preserving user privacy and preventing unauthorized entry.

4.1 ALGORITHMS USED

1. Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography is an advanced public-key cryptographic algorithm used to provide secure communication and identity verification. It is based on mathematical operations performed on elliptic curves over finite fields, which makes it highly secure even with smaller key sizes compared to traditional encryption methods.

In this project, ECC is used for generating cryptographic key pairs during user registration and authentication. The public key is used for encryption and verification, while the private key is securely stored and used for decryption and authentication validation. ECC helps in protecting authentication credentials and establishing trusted communication channels between users and cloud servers. Its efficiency and strong security properties make it suitable for cloud-based environments where performance and privacy are critical.

2. ChaCha20 Encryption Algorithm

ChaCha20 is a modern symmetric encryption algorithm designed to provide fast and secure data encryption. It uses a stream cipher mechanism that generates a pseudo-random keystream to encrypt sensitive information, making it resistant to cryptographic attacks and suitable for high-performance systems.

In this project, ChaCha20 is used to encrypt user credentials, authentication tokens, and session data before transmitting them over the network. This ensures that even if communication channels are compromised, attackers cannot interpret the encrypted information. The algorithm is lightweight and efficient, which makes it ideal for real-time authentication processes and secure cloud communication.

3. Multi-Factor Authentication Mechanism

Decision tree Multi-Factor Authentication is a security approach that requires users to verify their identity using multiple independent authentication factors. These factors may include something the user knows (password), something the user has (authentication token), or something the system verifies (session or device validation).

In this project, the multi-factor authentication mechanism is implemented to strengthen access control and reduce the risk of unauthorized login attempts. The system first validates user credentials, then verifies cryptographic tokens, and finally confirms session integrity before granting access. This layered verification process significantly improves authentication reliability and protects against attacks such as brute-force attempts and identity impersonation.

4. SHA-256 Algorithm

SHA-256 (Secure Hash Algorithm-256) is a widely used cryptographic hashing algorithm that converts input data into a fixed-length 256-bit hash value. It is designed to provide strong data integrity and security by ensuring that even a small change in the input produces a completely different hash output. Unlike encryption algorithms, hashing is a one-way process, meaning the original data cannot be directly retrieved from the hash value.

In this project, SHA-256 is used to securely process user authentication credentials before storing them in the cloud database. When a user registers or updates login information, the system generates a SHA-256 hash of the password and stores only the hashed value instead of the actual password. During authentication, the entered password is hashed again and compared with the stored hash to verify user identity.

This approach significantly improves security by protecting sensitive authentication data from database breaches and unauthorized access. Even if attackers gain access to stored records, they cannot easily recover the original credentials. SHA-256 also helps maintain data integrity during authentication processes and ensures reliable verification of user information within the privacy-enhanced cloud authentication framework.

4.2 SAMPLE CODE

```
from django.shortcuts import render

from django.template import RequestContext

from django.contrib import messages

import pymysql
```

PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION

```
from django.http import HttpResponse

from django.core.files.storage import FileSystemStorage

import datetime

from datetime import date

import os

from ecies.utils import generate_eth_key, generate_key

from ecies import encrypt, decrypt

import os

import json

from base64 import b64encode

from Crypto.Cipher import ChaCha20

from Crypto.Random import get_random_bytes

import timeit

from hashlib import sha256

import matplotlib.pyplot as plt

import io

import base64

import numpy as np

global username, propose, existing

#function to generate public and private keys for ECC algorithm

def ECCKeys():

    if os.path.exists("pvt.key"):

        with open("pvt.key", 'rb') as f:

            private_key = f.read()
```

```
f.close()

with open("pri.key", 'rb') as f:

    public_key = f.read()

f.close()

private_key = private_key.decode()

public_key = public_key.decode()

else:

    secret_key = generate_eth_key()

    private_key = secret_key.to_hex() # hex string

    public_key = secret_key.public_key.to_hex()

    with open("pvt.key", 'wb') as f:

        f.write(private_key.encode())

    f.close()

    with open("pri.key", 'wb') as f:

        f.write(public_key.encode())

    f.close()

    return private_key, public_key


#ECC will encrypt data using plain text adn public key

def EccEncrypt(plainText, public_key):

    ecc_encrypt = encrypt(public_key, plainText)

    return ecc_encrypt


#ECC will decrypt data using private key and encrypted text
```

PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION

```
def EccDecrypt(encrypt, private_key):  
  
    ecc_decrypt = decrypt(private_key, encrypt)  
  
    return ecc_decrypt  
  
def UploadFileAction(request):  
  
    if request.method == 'POST':  
  
        global propose, existing, username  
  
        filedata = request.FILES['t1'].read()  
  
        filename = request.FILES['t1'].name  
  
        access = request.POST.get('t2', False)  
  
        start = timeit.default_timer()  
  
        private_key, public_key = ECCKeys()  
  
        encrypted = EccEncrypt(filedata, public_key)  
  
        end = timeit.default_timer()  
  
        existing = end - start  
  
        start = timeit.default_timer()  
  
        key = get_random_bytes(32)  
  
        cipher = ChaCha20.new(key=key)  
  
        ciphertext = cipher.encrypt(filedata)  
  
        end = timeit.default_timer()  
  
        propose = end - start  
  
        msg = "Error in file upload"  
  
        db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password =  
'root', database = 'Authentication',charset='utf8')
```

PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION

```
db_cursor = db_connection.cursor()

student_sql_query = "INSERT INTO share(owner,filename,access_type)
VALUES('"+username+"','"+filename+"','"+access+"')"

db_cursor.execute(student_sql_query)

db_connection.commit()

print(db_cursor.rowcount, "Record Inserted")

if db_cursor.rowcount == 1:

    with open("AuthenticationApp/static/files/"+filename, "wb") as file:

        file.write(encrypted)

    file.close()

    msg = "File successfully loaded to cloud"

    context= {'data': msg}

    return render(request, 'UploadFile.html', context)


def DownloadAction(request):

    if request.method == 'GET':

        filename = request.GET.get('name', False)

        with open("AuthenticationApp/static/files/"+filename, "rb") as file:

            content = file.read()

        file.close()

        private_key, public_key = ECCKeys()

        decrypted = EccDecrypt(content, private_key)

        response = HttpResponse(decrypted,content_type='application/force-download')

        response['Content-Disposition'] = 'attachment; filename='+filename
```

```
    return response

def Graph(request):

    if request.method == 'GET':

        global existing, propose

        labels = ['ECC Computation', 'CHACHA Computation']

        height = [existing, propose]

        bars = labels

        y_pos = np.arange(len(bars))

        plt.figure(figsize = (4, 3))

        plt.bar(y_pos, height)

        plt.xticks(y_pos, bars)

        plt.xlabel("Technique Names")

        plt.ylabel("Computation Time")

        plt.title("Propose & Extension Computation Time Graph")

        plt.xticks()

        plt.tight_layout()

        buf = io.BytesIO()

        plt.savefig(buf, format='png', bbox_inches='tight')

        plt.close()

        img_b64 = base64.b64encode(buf.getvalue()).decode()

        context= {'data': img_b64}

        return render(request, 'ViewGraph.html', context)
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

5.1 Dataset Description

1. User ID

This column represents the unique identifier assigned to each registered cloud user. It is used to distinguish user records within the authentication database while supporting anonymous mapping mechanisms during secure login procedures.

2. Anonymous Identity Token

This field stores the system-generated anonymous authentication token used to mask the real identity of users during access requests. It helps in preserving user privacy while enabling secure verification by cloud services.

3. Encrypted Password Hash

This column contains the securely encrypted representation of the user's password generated using cryptographic hashing and encryption techniques. It ensures that sensitive credentials are protected against unauthorized access or database compromise.

4. Cryptographic Public Key

This column stores the public key generated during the Elliptic Curve Cryptography (ECC) key generation process. It is used for secure encryption, authentication verification, and establishing trusted communication channels between users and cloud servers.

5. Authentication Factor Type

This field represents the type of authentication factor used during login, such as Password, One-Time Token, or Session Verification. It helps analyze the effectiveness and usage patterns of different multi-factor authentication mechanisms.

6. Login Attempt Status

This column records the outcome of authentication attempts, such as Successful Login, Failed Login, or Blocked Access. It is useful for monitoring authentication performance and detecting suspicious login behavior.

7. Session Duration

This field captures the time interval for which an authenticated user session remains active. It helps evaluate session management policies and ensures timely logout to prevent unauthorized session hijacking.

8. Access Privilege Level

This column indicates the authorization level assigned to authenticated users, such as Normal User, Administrator, or Restricted Access. It supports role-based access control and secure resource management within cloud environments.

9. Authentication Risk Score

This column provides a calculated numerical score representing the risk level associated with a login attempt. It is used to prioritize security checks and trigger additional verification steps when abnormal authentication patterns are detected.

10. Encrypted Communication Status

This field indicates whether secure encrypted communication was successfully established during authentication. It helps validate the reliability of data protection mechanisms during credential transmission.

11. Failed Attempt Count

This column tracks the number of consecutive failed login attempts for a particular user account. It assists in identifying brute-force attacks and enforcing account lockout policies for enhanced security.

12. Timestamp of Authentication

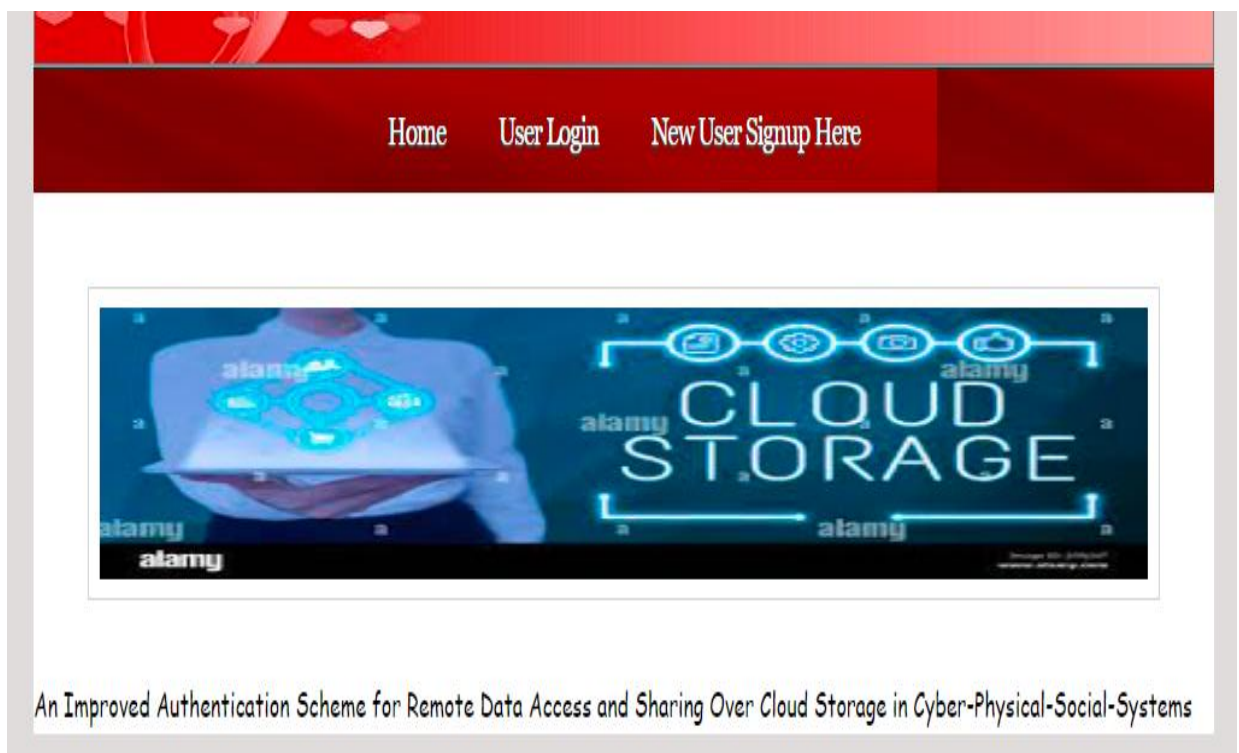
This field records the exact date and time of each authentication request processed by the system. It supports audit logging, forensic analysis, and tracking of user access patterns over time.

13. Security Alert Flag

This column stores a system-generated indicator that signals whether a login attempt triggered a security alert. It helps administrators quickly identify suspicious activities and respond proactively to potential authentication threats.

5.2 Result Analysis

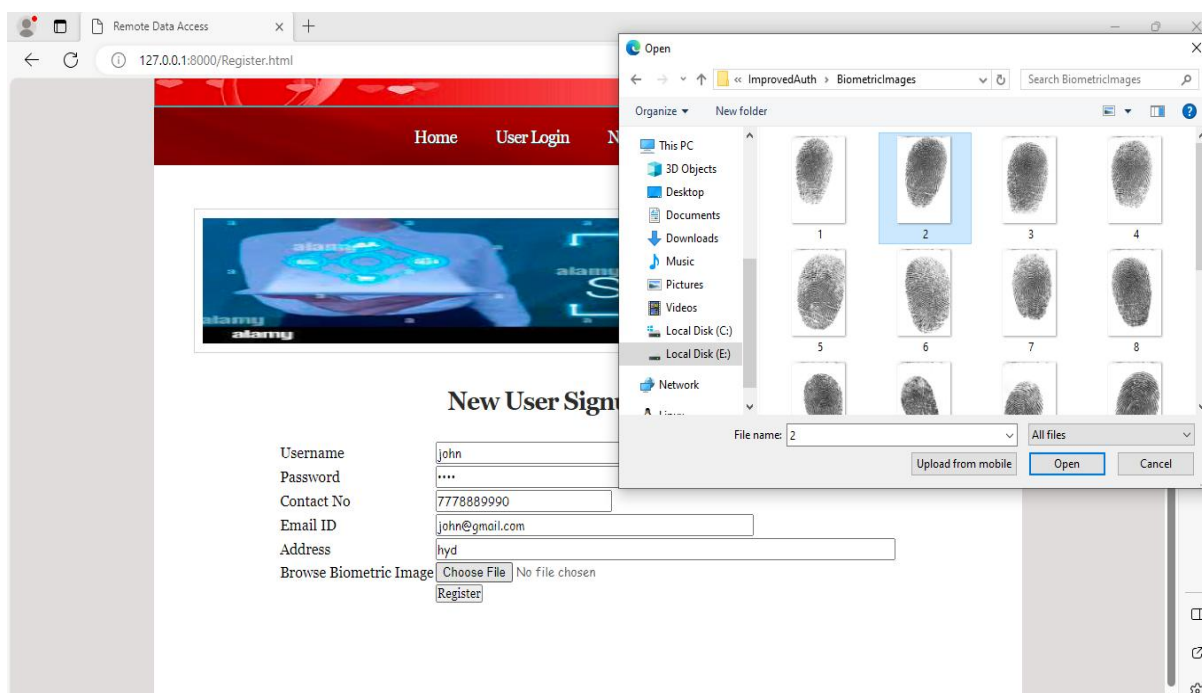
The Figure 5.1.illustrating the home page screen displays a website titled "Remote Data Access" with a red header featuring the text "An Improved Authentication Scheme for Remote Data Access and Sharing over Cloud Storage in Cyber-Physical." Below the header, there are navigation links. A central image showcases a person interacting with a holographic cloud storage interface, credited to Alamy. The page has a clean layout with a light gray background and a brief description of the authentication scheme below the image.



Screenshot 5.1: Home Page Screen

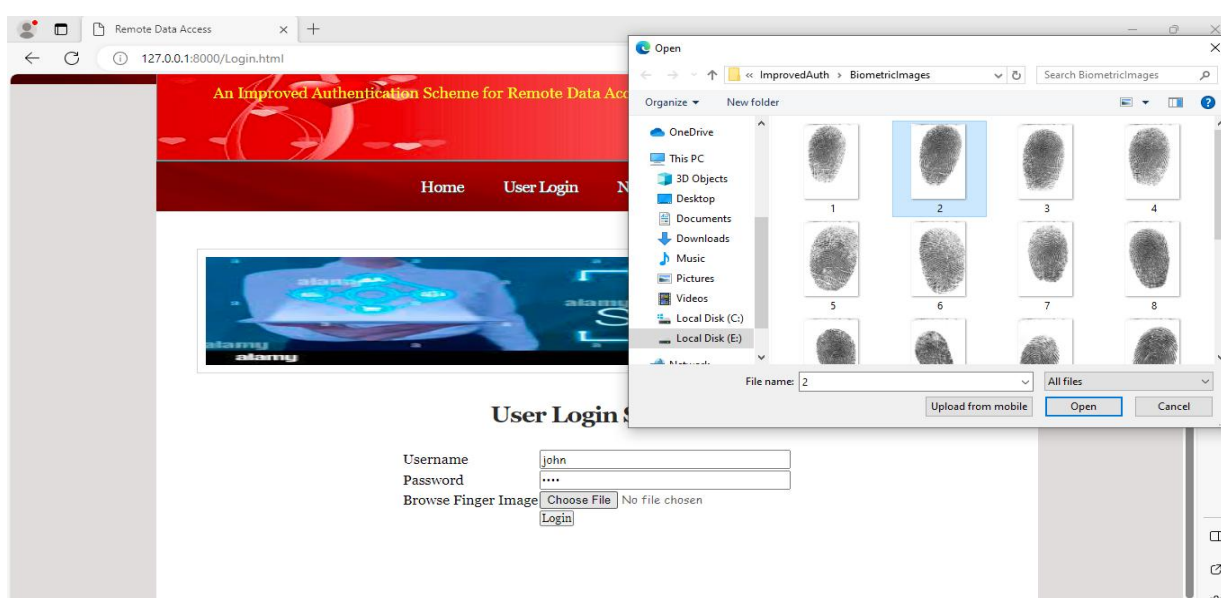
The screenshot 5.2 displays the new user signup screen features the same red header with the website title and navigation links. The main section includes a form for new users with fields for "Username," "Password," "Contact No," "Email ID," "Address," and "Browse Biometric Image" with a registration option. A file explorer window is open, displaying fingerprint images for selection, with one image highlighted. The background includes the same cloud storage image credited to Alamy.

PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION



Screenshot 5.2: New User Signup Screen.

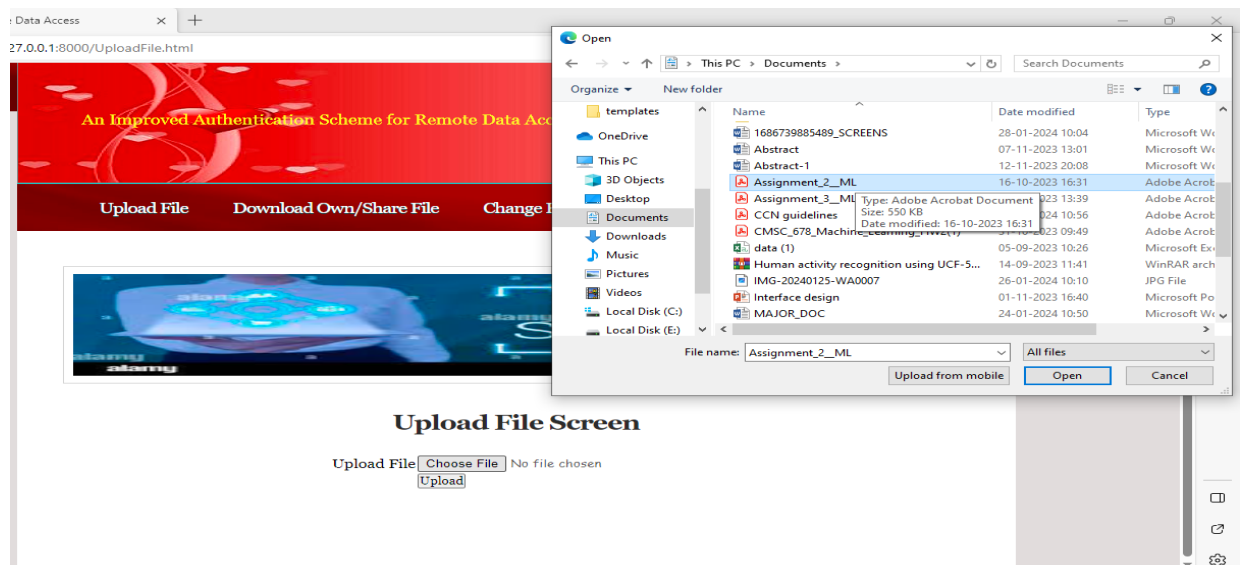
The screenshot 5.3 describes the user login screen has the red header with the website title and navigation links. The main content shows a simple login form with fields for "Username" (pre-filled as "john") and "Password" (partially obscured), along with a "Browse Finger Image" option. A file explorer window is open, displaying multiple fingerprint images, with one selected. The layout mirrors the previous screens with the Alamy cloud storage image and a light gray background.



Screenshot 5.3: User Login Screen.

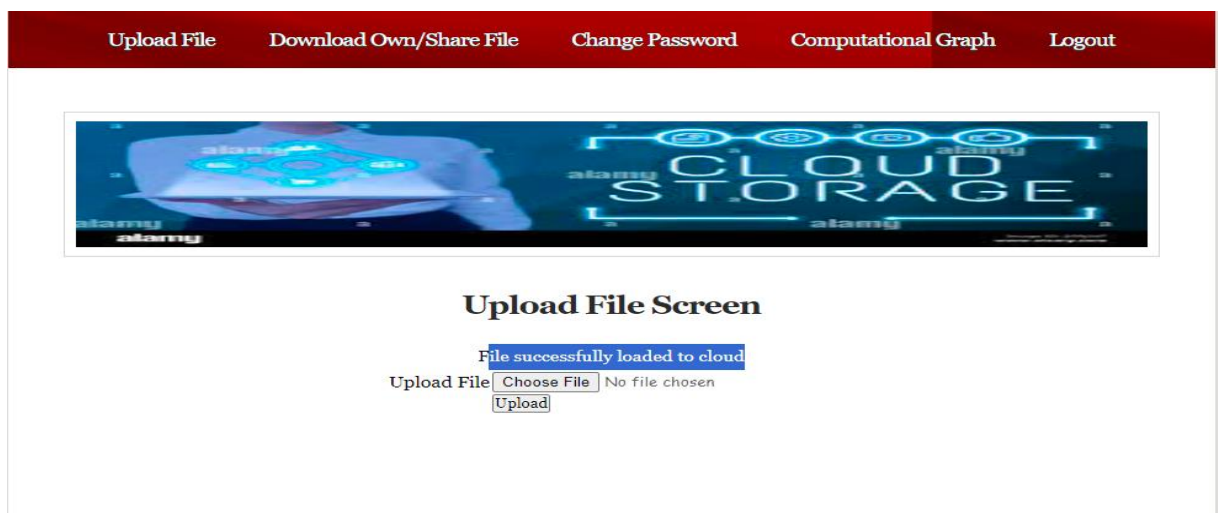
PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION

Figure 5.4 illustrates the upload file screen features the red header with the website title and navigation links. The central section includes a "Upload File Screen" title and a file selection option, with a file explorer window open showing various document files. The familiar cloud storage image from Alamy is present, and the page maintains the light gray background typical of the site.



Screenshot 5.4: Upload File Screen.

The Screenshot 5.5 shows the file successfully uploaded screen retains the red header with the website title and navigation links. The main section displays a message "File successfully loaded to cloud" followed by a file upload option. The Alamy cloud storage image remains central, and the light gray background continues, indicating a successful file upload process



Screenshot 5.5 File Successfully Uploaded Screen.

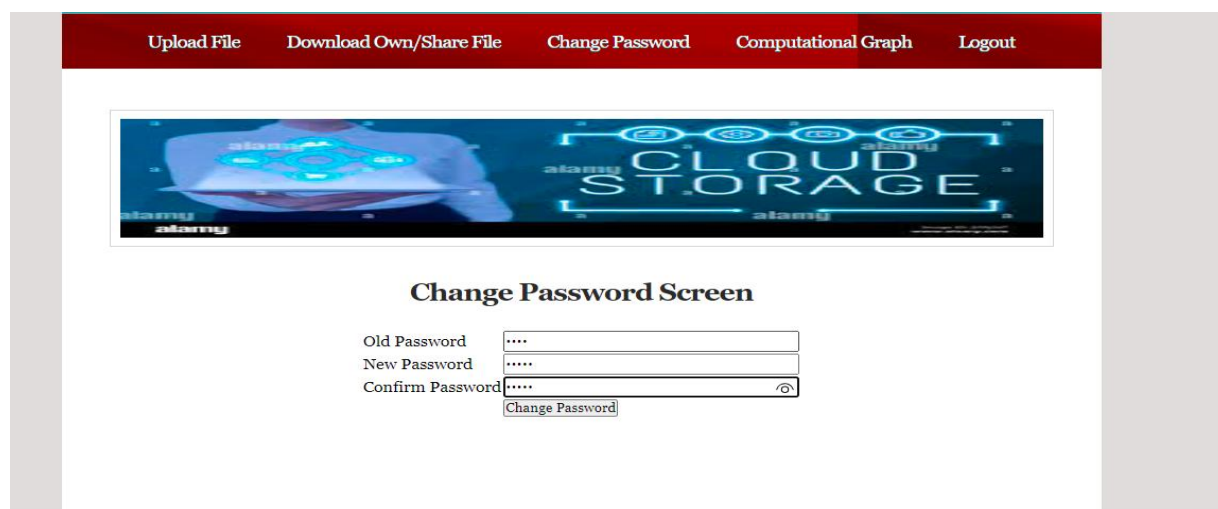
PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION

The Screenshot 5.6 illustrates a website titled "Remote Data Access" with a red header featuring the text "An Improved Authentication Scheme for Remote Data Access and Sharing Over Cloud Storage in Cyber-Physical." It displays navigation links and a central image of a person with a holographic cloud storage interface credited to Alamy. The main section shows a table listing files with columns for "Owner Name," "File Name," and "Download File,".



Screenshot 5.6: Download Screen

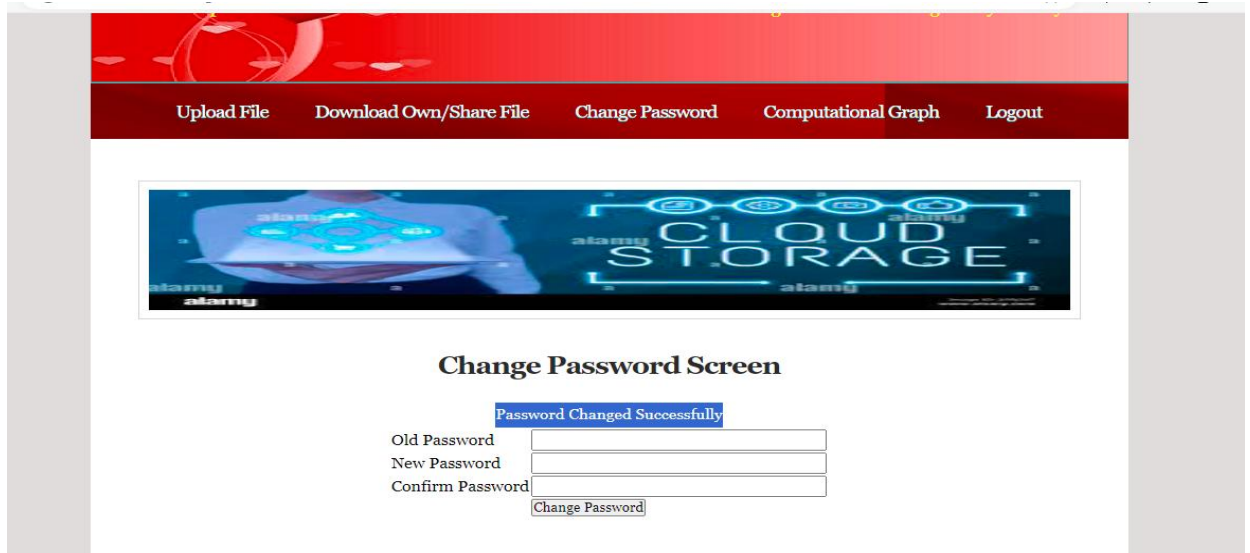
The Screenshot 5.7 describes a website with the same red header and title, along with navigation links and the Alamy cloud storage image. It shows a "Change Password Screen" section with fields labeled "Old Password," "New Password," and "Confirm Password," indicating a password update process on a light gray background.



Screenshot 5.7: Change Password Screen.

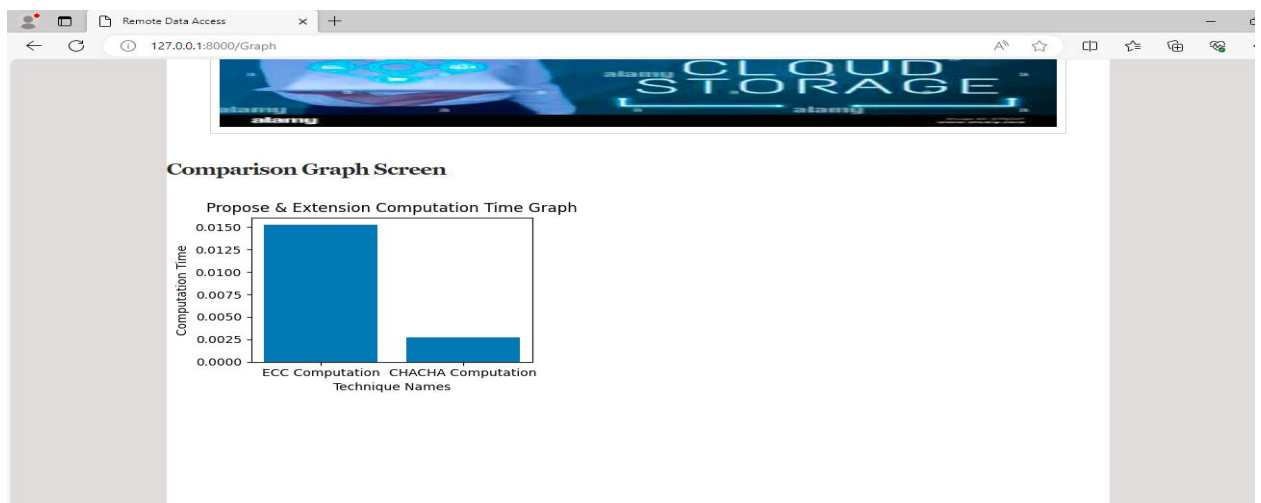
PRIVACY-ENHANCED ANONYMOUS CLOUD AUTHENTICATION FRAMEWORK WITH MULTI-FACTOR PROTECTION

The figure 5.8 displays the website with the red header, navigation links, and the Alamy cloud storage image. It shows a "Change Password Screen" section with a "Password Changed Successfully" message, followed by fields for "Old Password," "New Password," and "Confirm Password," suggesting a completed password update on the light gray background.



Screenshot 5.8: Successfully Password Changed.

Screenshot 5.9 shows a website with the red header, navigation links, and the Alamy cloud storage image. It displays a "Comparison Graph Screen" section with a bar graph titled "Propose & Extension Computation Time Graph," comparing computation times for "ECC Computation" and "CHACHA Computation" techniques, with values around 0.015 and 0.0025 respectively, presented on a light gray background.



Screenshot 5.9: Model Comparison Screen.

6. VALIDATION

6. VALIDATION

6.1 INTRODUCTION

Validation is a crucial phase in the development of the *Privacy-Enhanced Anonymous Cloud Authentication Framework with Multi-Factor Protection*, ensuring that the authentication system operates securely, efficiently, and reliably under different usage scenarios. The primary objective of validation is to confirm that the system satisfies all functional and security requirements while delivering correct authentication outcomes for various input conditions.

In this project, validation mainly focuses on testing the framework's capability to manage real-world cloud authentication operations involving multiple users, encrypted credentials, and anonymous identity verification. The system is evaluated for its ability to handle secure user registration, encrypted data storage, anonymous credential generation, and multi-factor authentication verification without compromising performance or privacy.

Additionally, validation ensures that cryptographic mechanisms such as secure key generation, encrypted communication, and session validation work correctly across different modules. The system is also tested for resilience against invalid inputs, unauthorized access attempts, and network interruptions during authentication procedures. Overall, validation confirms that the proposed framework can provide reliable privacy-preserving authentication and maintain secure access control in cloud computing environments.

6.2 TEST CASES

Test cases are designed to evaluate the functionality, performance, usability, and security of the authentication framework. Each test case defines a specific objective, input scenario, expected system response, and observed result. The test cases are organized based on different system modules to ensure comprehensive validation of all authentication processes.

6.2.1 UPLOADING DATASET

This module validates the system's ability to securely register users, generate anonymous identities, and store encrypted credentials in the cloud database. It ensures that valid inputs are accepted, invalid data is rejected, and sensitive authentication details are properly protected.

Test Cases

Test Case ID	Test Case Name	Objective	Input	Expected Output	Result
TC_01	Valid User Registration	Verify successful registration	Valid user details	User registered successfully	Pass
TC_02	Empty Field Validation	Check handling of missing data	Blank input fields	Error message displayed	Pass
TC_03	Duplicate User Entry	Validate duplicate prevention	Existing username/email	Registration rejected	Pass
TC_04	Encrypted Credential Storage	Test encryption process	Valid password	Credentials stored securely	Pass
TC_05	Invalid Input Format	Validate data validation rules	Incorrect email/format	Input rejected	Pass
TC_06	Anonymous Identity Generation	Verify privacy mechanism	Valid registration request	Anonymous ID generated	Pass
TC_07	Special Character Handling	Check input preprocessing	Symbols in username	Accepted or sanitized	Pass

6.2.2 CLASSIFICATION

This module validates the system's ability to perform secure authentication using multiple verification factors such as password validation, cryptographic token verification, and session confirmation. It ensures that authentication decisions are accurate, timely, and resistant to unauthorized access attempts.

Test Cases

Test Case ID	Test Case Name	Objective	Input	Expected Output
TC_01	Valid Login Authentication	Verify successful login	Correct credentials	Access granted
TC_02	Incorrect Password	Validate failure handling	Wrong password	Access denied
TC_03	Token Verification Failure	Test multi-factor validation	Invalid authentication token	Login rejected
TC_04	Session Timeout Handling	Check session security	Idle authenticated session	Automatic logout
TC_05	Unauthorized Access Attempt	Validate security robustness	Unknown user request	Access blocked
TC_06	Replay Attack Simulation	Test attack resistance	Reused authentication request	Request rejected
TC_07	Multiple Login Requests	Validate concurrent handling	Multiple users login	All requests processed

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

7.1 Project Conclusion

The project *Privacy-Enhanced Anonymous Cloud Authentication Framework with Multi-Factor Protection* presents a secure and intelligent authentication mechanism designed to protect user identity and sensitive data in cloud computing environments. The system focuses on strengthening access control by integrating privacy-preserving authentication techniques with multiple security verification layers.

The proposed framework performs several stages including secure user registration, anonymous identity generation, encrypted credential storage, and multi-factor authentication verification. Sensitive authentication data is protected using advanced cryptographic techniques such as Elliptic Curve Cryptography and ChaCha20 encryption, which ensure confidentiality, integrity, and resistance against common cyber-attacks. Anonymous credential generation helps in preventing identity leakage while maintaining traceability for authorized cloud services.

The framework also incorporates multiple authentication factors such as password verification, cryptographic token validation, and session-based authentication mechanisms to enhance overall system security. These mechanisms collectively reduce the risk of unauthorized access, impersonation attacks, replay attacks, and credential theft. Additionally, secure communication channels are established between users and cloud servers to ensure protected data transmission during authentication processes.

The developed system demonstrates that privacy-enhanced authentication models can significantly improve trust and security in cloud platforms. By replacing traditional single-factor login systems with intelligent multi-factor and anonymous authentication techniques, the framework provides stronger protection, improved user privacy, and reliable identity verification. Overall, the project contributes to building more secure and privacy-aware cloud infrastructures by enabling proactive defense against authentication-related threats.

7.2 Future Aspects

Although the proposed authentication framework achieves strong security and privacy protection, several enhancements can be implemented in future work to further improve its performance and scalability.

In the future, advanced biometric authentication techniques such as facial recognition, fingerprint verification, or behavioral biometrics can be integrated to provide more robust and user-friendly authentication experiences. The modular architecture of the system allows seamless integration of such additional authentication factors without major structural modifications.

Another potential improvement is the incorporation of blockchain-based identity management for decentralized authentication and tamper-proof logging of access activities. This would enhance transparency, eliminate reliance on centralized identity providers, and strengthen trust among cloud users and service providers.

The framework can also be enhanced by implementing AI-driven anomaly detection models that monitor login patterns and detect suspicious authentication attempts in real time. Using larger datasets of authentication logs and cyber-attack patterns could improve the system's ability to identify sophisticated threats such as credential stuffing or insider attacks.

Future research may also focus on developing a comprehensive cloud security dashboard where administrators can visualize authentication attempts, threat alerts, access histories, and risk levels dynamically. Such visualization tools would support faster decision-making and efficient incident response in enterprise cloud environments.

Finally, the proposed framework can be extended to support authentication in emerging technologies such as Internet of Things (IoT), edge computing, and hybrid cloud systems, where secure and privacy-preserving access control mechanisms are increasingly required. This expansion would further strengthen cybersecurity readiness across multiple digital ecosystems.

8. BIBLIOGRAPHY

8. BIBLIOGRAPHY

8.1 References

- [1] Mostafa, A.M.; Ezz, M.; Elbashir, M.K.; Alruily, M.; Hamouda, E.; Alsarhani, M.; Said, W. Strengthening Cloud Security: An Innovative Multi-Factor Multi-Layer Authentication Framework for Cloud User Authentication. *Appl. Sci.* **2023**, *13*, 10871. <https://doi.org/10.3390/app131910871>
- [2] Alatawi, M.N. Blockchain-Driven Smart Contracts for Advanced Authorization and Authentication in Cloud Security. *Electronics* **2025**, *14*, 3104. <https://doi.org/10.3390/electronics14153104>
- [3] Tanveer, M.; Badshah, A.; Khan, A.; Alasmay, H.; Chaudhry, S. CMAF-IIoT: Chaotic map-based authentication framework for industrial internet of things. *Internet Things* **2023**, *23*, 100902.
- [4] Alasmay, H.; Tanveer, M. ESCI-AKA: Enabling secure communication in an iot-enabled smart home environment using authenticated key agreement framework. *Mathematics* **2023**, *11*, 3450.
- [5] Qui, S.; Wang, D.; Xu, G.; Kumari, S. Practical and provably secure three-factor authentication protocol based on extended chaotic maps for mobile lightweight devices. *IEEE Trans. Dependable Secur. Comp.* **2022**, *20*, 1338–1351.
- [6] Alsirhani, A.; Ezz, M.; Mostafa, A.M. advanced authentication mechanisms for identity and access management in cloud computing. *Comp. Syst. Sci. Eng.* **2022**, *43*, 967–984.
- [7] Wang, Q.; Wang, D.; Cheng, C.; He, D. Quantum2FA: Efficient quantum-resistant two-factor authentication scheme for mobile devices. *IEEE Trans. Dependable Secur. Comp.* **2021**, *20*, 193–208.
- [8] Wu, L.; Wen, Y.; Yi, J. A Higher Performance Data Backup Scheme Based on Multi-Factor Authentication. *Entropy* **2024**, *26*, 667. <https://doi.org/10.3390/e26080667>
- [9] Hu, H.; Lin, C.; Chang, C.C.; Chen, L. Enhanced secure data backup scheme using multi-factor authentication. *IET Inf. Secur.* **2019**, *13*, 649–658.
- [10] Yi, J.; Wen, Y. An Improved Data Backup Scheme Based on Multi-Factor Authentication. In Proceedings of the 9th International Conference on Big Data Security on Cloud (BigDataSecurity), IEEE International Conference on High Performance and Smart

Computing, (HPSC) and IEEE International Conference on Intelligent Data and Security (IDS), New York, NY, USA, 6–8 May 2023.

- [11] Chang, C.C.; Chou, Y.C.; Sun, C.Y. Novel and practical scheme based on secret sharing for laptop data protection. *IET Inf. Secur.* **2015**, *9*, 100–107.
- [12] Bamashmos, S.; Chilamkurti, N.; Shahraki, A.S. Two-Layered Multi-Factor Authentication Using Decentralized Blockchain in an IoT Environment. *Sensors* **2024**, *24*, 3575. <https://doi.org/10.3390/s24113575>
- [13] Wu, Y.; Pang, M.; Ma, J.; Ou, W.; Yue, Q.; Han, W. An Identity Management Scheme Based on Multi-Factor Authentication and Dynamic Trust Evaluation for Telemedicine. *Sensors* **2025**, *25*, 2118. <https://doi.org/10.3390/s25072118>

8.2 Github Link

<https://github.com/Pravallikapalvai/privacy-enhanced-anonymous-cloud-authentication-framework-with-multi-factor-protection>